

MARCOS AGUILERA-KEYSER

APPLIED ML ENGINEERING | GENERATIVE AI & RAG SYSTEMS

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Hands-on ML professional with extensive experience implementing predictive models and enterprise AI systems across SAS, Liberty Mutual, and New York Life. Since 2023, contributed to production RAG and agentic QA pipelines and then to performance-improvement work in a Generative AI Center of Excellence. Combines Python engineering with model adaptation, retrieval, and evaluation expertise.

ENGINEERING FOCUS

End-to-end ML and RAG pipelines; retrieval and model integration; evaluation tooling; reproducible experiments; domain adaptation; collaboration across data science, ML engineering, and MLOps.

PROFESSIONAL EXPERIENCE

Principal Applied Research Scientist - Generative AI

New York Life | Jan 2023 - Jul 2026

- Contributed within a five-to-six-person team to production RAG systems implemented directly in Python, covering end-to-end QA pipelines and evaluation without relying on orchestration frameworks such as LangChain.
- Supported the evolution of retrieval-augmented QA into agentic workflows, working across system implementation, model behaviour, and evaluation.
- In the Generative AI CoE, investigated production performance problems and developed retrieval, model-adaptation, and architectural interventions for insurance procedures.
- Built reproducible retrieval workflows with corpus preparation, relevance-label alignment, duplicate and leakage checks, lexical and dense model evaluation, embedding caches, and versioned pipeline stages.
- Implemented domain-adaptation experiments for encoders and LLMs, including MiniLM training and QLoRA adaptation of Llama 3.1 8B within a single-GPU compute budget.
- Built a procedural-RAG prototype combining neural and knowledge-graph approaches, together with evaluation components for missing steps and procedural structure; taught colleagues retrieval and agent-learning techniques.

Lead Data Scientist / Senior Data Scientist

New York Life | 2017 - Dec 2022; Lead role Jan 2021 - Dec 2022

- Engineered media mix modelling workflows for marketing allocation, with a 4-12% increase in weekly attributed sales.
- Implemented spatial ML and PyTorch LSTM pipelines on AWS for travel-time prediction and agent-area planning, including geospatial indexing across 2.5 million hexagons.
- Built speech analytics, transcription, and web-data collection workflows; established coding standards and reproducible ML project practices for cross-functional delivery.

EXPERIENCE, CONTINUED

Founding Machine Learning Engineer

Dejake | 2016 - 2017

- Developed neural sequence models and LSTM/RNN pipelines for e-commerce intent recognition, automated dialogue, and cart-recovery applications.

Assistant Director, Data Science - Solaria Labs

Liberty Mutual | 2013 - 2016

- Developed and deployed machine learning services for catastrophe severity modelling across Property and Auto portfolios, including event- and peril-level severity estimates.
- Built NLP feature-engineering services and sentiment-analysis pipelines for catastrophe-related policyholder text.

Machine Learning Software Engineer - SAS Enterprise Miner R&D

SAS Institute | 2007 - 2013

- Contributed algorithms and software to enterprise analytics products, spanning data mining, predictive modelling, and text-processing workflows.
- Implemented text-mining and statistical ML methods including TF-IDF, latent semantic analysis, rule-based parsing, and density-based clustering.

TECHNICAL EDUCATION & RECOGNITION

Colorless Green: Created a public retrieval course covering lexical and neural search, hybrid architectures, domain adaptation, and evaluation. colorlessgreen.ai

NLP & Machine Learning Instructor | Principal Analytics Prep., New York | Part-time, 2018 - 2020

Kaggle competition winner: Liberty Mutual Group - Fire Peril Loss Cost (2014).

EDUCATION

Graduate studies in Economics | University of Oxford & Louvain University

Advanced graduate coursework in Financial Economics.

B.S. in Economics | Carlos III University & Louvain University

SELECTED TECHNICAL SKILLS

Languages & delivery: Python, SQL, Bash, object-oriented programming, Git, GitHub Actions, Docker, DVC, Terraform, Linux, Python packaging, testing and reproducible pipelines.

ML & retrieval: PyTorch, Hugging Face Transformers, LoRA/QLoRA, bi-encoders, cross-encoders, BM25, hybrid retrieval, knowledge graphs, RAG and agent evaluation.

Platforms & observability: AWS Bedrock, SageMaker, OpenSearch, Lambda, Textract; Weights & Biases, LangSmith, RAGAS; ART and agent-learning workflows.